

How Do We Know Our Intake Is Doing a Good Job?

Measurement is the cornerstone of professional trust in automated legal systems. Before a *Fachanwalt für Arbeitsrecht* can adopt technology to handle sensitive documents like termination notices (*Kündigungsschreiben*) or employment contracts (*Arbeitsvertrag*), that technology must prove it can "read" and "reason" with the same precision as a human expert. We do not rely on the "hallucinations" of a generic AI; our system is a deterministic engine built on German regexes and keyword scoring. We subject it to a rigorous evaluation framework—essentially a high-stakes school exam for software.

1. The Starting Story: The Folder of Trust

Imagine a father sitting with his curious child. He hands the child a thick folder full of important papers. "If I give this folder to someone and ask them to help me," the father explains, "I need to know they aren't just looking at the pictures. I need to know they truly understand what is happening to our family's future."

To trust an automated system with that same folder, we must answer six fundamental "Trust Questions":

1. **Did we read it correctly?** (Accuracy of characters and numbers.)
2. **Did we understand the documents?** (Correct classification of document types.)
3. **Did we find the important facts?** (Extraction of dates and names with provenance.)
4. **Did those facts stay correct?** (Preservation of truth as data moves through layers.)
5. **Was the case built correctly?** (Structural consistency and legal safety rules.)
6. **Did the whole journey produce the expected result?** (The final end-to-end outcome.)

Answering these questions is how we move from blind faith to verified, architectural trust.

2. The Fairness Principle: The Teacher and the Student

To test a student fairly, the teacher must have an answer key prepared *before* the student sees the exam. In our framework, our lawyers act as the teachers. They prepare a "Golden YAML"—a master answer sheet representing the absolute ground truth of a case, independent of the software.

The strategic principle is absolute: **Intake never creates the answers used to judge itself.** By using independent verification, we prevent "circular logic" in our testing. If the system were allowed to influence its own benchmark, the results would be meaningless for risk management and *Rechtsdienstleistungsgesetz* (RDG) compliance.

3. The Quality Journey: Six Stages of Verification

Our software processes a case through a modular architecture of six stages (L0–L5). This modularity is a critical architectural differentiator; it allows us to upgrade the OCR (READ) layer without risking the integrity of the Legal Logic (CHECK) layer. We test these stages in isolation to ensure that a simple reading error doesn't unfairly make our complex logic look broken.

- **READ (L0):** Accuracy of words and numbers. We employ a "fail-closed" rule: if a document is unreadable, the system stops rather than guessing.
- **UNDERSTAND (L1):** Identifying document types (e.g., *Abmahnung* vs. *Kündigung*) without wild guessing.
- **FIND (L2):** Fact extraction. The system identifies data points and connects them to the evidence on the page using an "RDG-safe" lexicon.
- **ORGANISE (L3):** Placing facts into the case file. If documents disagree, the system preserves the conflict for human resolution.
- **CHECK (L4):** Verifying structural consistency against the *Kündigungsschutzgesetz* (KSchG) and *BetrVG* rules.
- **VERIFY (L5):** The final end-to-end result—the bottom line for the legal professional.

4. Integrity of the Test: Safety Invariants

In our architectural methodology, we distinguish between **Product Quality** and **Test Execution Quality**. If the teacher's answer sheet is smudged, a "zero" score is a test error, not a student error. We use seven "Safety Invariants" (INV-01 to INV-07) to maintain this integrity.

Crucially, our system is designed to **fail-closed**. For instance, under INV-03, the absence of evidence never creates a "MISSING" status unless explicitly attested by a human in a signed *Nachweisübersicht*. This "Hermeneutic Boundary Guard" ensures that the machine never oversteps its role as a recorder of facts. Only valid, perfectly scoreable tests where the "Golden" data is perfect are allowed to influence our public performance reporting.

5. Intake Quality Report: Actual Performance Results

The following data represents our "Golden-2026-09-25" baseline, reflecting 240 jobs across 40 lawyer-verified cases. Note that performance varies by route complexity; Route A (Simple) currently performs at 71.4% classification accuracy, while Route D (Public Sector/Personalrat) represents a more complex logic set currently under optimization at 51.5%.

Quality Question	Actual Result (n/d)	What It Means
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Did we read correctly?	100% (404/404)	Perfect recovery of characters and numbers from critical text.
Did we understand correctly?	59.2% (197/333)	Correct identification of document type in 6 out of 10 cases.
Did dangerous mistakes occur?	21.2% (53/250)	The "Wrong-Confident" rate; how often the system was sure but wrong (Lower is better).
Did we find the right facts?	42.9% / 55.1%	Fact Recall (188/438) vs. Precision (188/341).
Did correct facts stay correct?	84.0% (368/438)	Preservation of truths across architectural layers.
Was the case properly built?	82.5% (33/40)	How often the final file met all technical and legal structure rules.

6. The Three Most Important Final Results

We synthesize our performance into three high-level indicators that determine commercial and legal readiness:

VERIFIED CASE PASS RATE (Metric 25): 2.5%

Representing 1 out of 40 cases, this is our "strict" pass rate for the complete journey. It requires every Tier-1 value and legal state to be 100% correct from scan to export.

CRITICAL FIELD ACCURACY (Metric 26): 63.9%

For Tier-1 data—the information that matters most for deadlines and gates (235/368)—the system is correct nearly two-thirds of the time.

CRITICAL ERROR ESCAPE RATE (Metric 27): 63.0%

LOWER IS BETTER. This measures how often a serious machine mistake (133/211) was not caught by the scripted review before the final exit. Reducing this number is the primary focus of our current engineering cycle.

Note: The Human Correction Burden (Metric 28) is not measured in the current available results and requires a sanctioned human study.

7. Forensic Improvement: Traceability

When the system fails, we do not simply acknowledge the error; we perform a "Root Cause Analysis" through total traceability. Every fact in the system has a "provenance"—a direct link back to its source.

If a date is wrong in the final case file, our engineers "trace" the error back through the layers: Did we **READ** the character wrong? Did the **FIND** layer extract the wrong number? Or did the **ORGANISE** step overwrite a confirmed value? Because our system is deterministic and uses no LLM "black box," we can see exactly which German sentence in which PDF caused the failure. These failures are maps that tell us exactly which gear in the machine needs tightening.

8. The Message: Why We Watch the Machine

At the end of the day, our philosophy remains "Trust but Verify."

The child in our story asks: "Do we just trust the computer to do it?"

The father explains: "No. We know what the correct answer should be because we are the experts. We watch every important step. We compare the results independently. We measure what went well so we can fix what didn't. That is how we know it is doing a good job."